

FINAL

Angad Kapoor
005-697-193

Problem #1

a) We have $p_{k+1} = \sqrt{9 - p_k}$, $p_0 \in [0, 9]$

$$\Rightarrow g(x) = \sqrt{9 - x}, \quad [a, b] = [0, 9]$$

The function $g(x)$ is continuous on the interval $(-\infty, 9]$

$$\therefore g(x) \in C[a, b]$$

$$g'(x) = \frac{1}{2\sqrt{9-x}} \cdot -1 = \frac{-1}{2\sqrt{9-x}}$$

$g'(x)$ is continuous and well-defined on the interval $(-\infty, 9)$.

Since $2\sqrt{9-x} > 0$ when $x \in (-\infty, 9)$, we know that $g'(x) < 0 \quad \forall x \in [a, b]$.

$\hookrightarrow$ we can see that $\lim_{x \rightarrow 9^-} g'(x) = -\infty < 0$,

so on the given interval $[0, 9]$, $g(x)$ is strictly decreasing.

$$\Rightarrow g(a) \geq g(p) \geq g(b) \quad \forall p \in [a, b]$$

$$g(a) = g(0) = \sqrt{9-0} = 3 \in [0, 9]$$

$$g(b) = g(9) = \sqrt{9-9} = 0 \in [0, 9]$$

$\therefore$ since g is strictly decreasing on the given interval, and $g(a) \in [a, b]$ and $g(b) \in [a, b]$ we can conclude that the corresponding fixed-point method is well-defined.

$$\therefore g: [0, 9] \longrightarrow [0, 9] \quad \checkmark$$

b) According to the Fixed-Point Method convergence theorem:

For a function $g(x)$, if the following criteria are satisfied:

- $g \in C[a, b]$, $g' \in C[a, b]$
- $g: [a, b] \mapsto [a, b]$
- $|g'(x)| < K$, $K \in (0, 1) \quad \forall x \in [a, b]$

then the FPM converges to a unique fixed point $p^* \in [a, b]$.

We have $g(x) = \sqrt{9-x}$, $[a, b] = [0, 9]$

$$\therefore |g'(x)| = \left| \frac{-1}{2\sqrt{9-x}} \right| = \frac{1}{2\sqrt{9-x}}$$

We can see that $|g'(x)| = 1$ when $x = 8.75$,
but $|g'(x)| > 1$ when $x > 8.75$.

thus the interval $[0, 9]$ must be partitioned into two intervals: $[0, 8.75) \cup [8.75, 9]$

Consider the first interval: $[0, 8.75)$

$$\begin{aligned} &\rightarrow g(x), g'(x) \in C[0, 9] \\ &\therefore g(x), g'(x) \in C[0, 8.75) \quad \checkmark \end{aligned}$$

$\rightarrow$ we saw that g is strictly decreasing on the interval $[0, 9]$, so g is strictly decreasing on $[0, 8.75)$

$$\begin{aligned} g(0) &= 3 \in [0, 8.75) \\ g(8.75) &= 0.5 \in [0, 8.75) \end{aligned}$$

$$\therefore g: [0, 8.75) \mapsto [0, 8.75) \quad \checkmark$$

$$\rightarrow |g'(x)| < 1 \quad \forall x \in [0, 8.75) \quad \checkmark$$

$\therefore$ the FPM converges for any starting point p_0 ,
 $p_0 \in [0, 8.75)$

Now, for the second interval: $[8.75, 9]$

we see that $|g'(x)| \geq 1$, so we cannot apply the FPM convergence theorem. However, we can observe that any starting point $p_0 \in [8.75, 9]$ eventually returns to the interval $[0, 8.75]$:

$$g(8.75) = \frac{1}{2} \in [0, 8.75)$$

$$g(9) = 0 \in [0, 8.75)$$

$$\therefore g: [8.75, 9] \longrightarrow [0, 8.75)$$

Now, we have already proved that the FPM convergence theorem holds for $[0, 8.75)$, so we can reason that any starting point $p_0 \in [8.75, 9]$ will map into $[0, 8.75)$, and then converge.

$\therefore$ the FPM converges to a unique fixed point $\forall x \in [0, 9]$ ✓

Problem #2

To show that this formula holds, we can apply Taylor's Theorem to the RHS.

$$\text{Given: } \frac{\partial^2 f}{\partial x \partial y} = \frac{1}{2h^2} \left(\overset{(A)}{f(x+h, y+h)} - \overset{(B)}{f(x+h, y)} - \overset{(C)}{f(x, y+h)} + \overset{(D)}{2f(x, y)} \right. \\ \left. - \overset{(E)}{f(x-h, y)} - \overset{(F)}{f(x, y-h)} + \overset{(G)}{f(x-h, y-h)} \right) + \Theta(h^2)$$

$$\text{for } f \in C^4([x-h, x+h] \times [y-h, y+h])$$

Using the method of undetermined coefficients, let:

$$\frac{\partial^2 f}{\partial x \partial y} = Af(x+h, y+h) + Bf(x+h, y) + Cf(x, y+h) \\ + Df(x, y) + Ef(x-h, y) + Ff(x, y-h) \\ + Gf(x-h, y-h) + \Theta(h^2)$$

• we have the general formula, by Taylor's Theorem:

$$f(x+a, y+b) = f(x, y) + a \frac{\partial f}{\partial x}(x, y) + b \frac{\partial f}{\partial y}(x, y) \\ + \frac{a^2}{2} \frac{\partial^2 f}{\partial x^2}(x, y) + ab \frac{\partial^2 f}{\partial x \partial y}(x, y) + \frac{b^2}{2} \frac{\partial^2 f}{\partial y^2}(x, y) \\ + \frac{a^3}{6} f_{xxx}(x, y) + \frac{a^2 b}{2} f_{xxy}(x, y) + \frac{ab^2}{2} f_{xyy}(x, y) + \frac{b^3}{6} f_{yyy}(x, y) \\ + \Theta(a^4 + b^4)$$

Therefore:

$$\rightarrow f(x+h, y+h) = f(x, y) + h(f_x(x, y) + f_y(x, y)) \\ + \frac{h^2}{2}(f_{xx}(x, y) + 2f_{xy}(x, y) + f_{yy}(x, y)) \\ + \frac{h^3}{6}(f_{xxx}(x, y) + 3f_{xxy}(x, y) + 3f_{xyy}(x, y) + f_{yyy}(x, y)) \\ + \Theta(h^4)$$

$$\rightarrow f(x+h, y) = f(x, y) + hf_x(x, y) + \frac{h^2}{2} f_{xx}(x, y) \\ + \frac{h^3}{6} f_{xxx}(x, y) + \Theta(h^4)$$

$$\rightarrow f(x, y+h) = f(x, y) + hf_y(x, y) + \frac{h^2}{2} f_{yy}(x, y) + \frac{h^3}{6} f_{yyy}(x, y) + \theta(h^4)$$

$$\rightarrow f(x-h, y) = f(x, y) - hf_x(x, y) + \frac{h^2}{2} f_{xx}(x, y) - \frac{h^3}{6} f_{xxx}(x, y) + \theta(h^4)$$

$$\rightarrow f(x, y-h) = f(x, y) - hf_y(x, y) + \frac{h^2}{2} f_{yy}(x, y) - \frac{h^3}{6} f_{yyy}(x, y) + \theta(h^4)$$

$$\begin{aligned} \rightarrow f(x-h, y-h) &= f(x, y) - h(f_x(x, y) + f_y(x, y)) \\ &\quad + \frac{h^2}{2}(f_{xx}(x, y) + 2f_{xy}(x, y) + f_{yy}(x, y)) \\ &\quad - \frac{h^3}{6}(f_{xxx}(x, y) + 3f_{xxy}(x, y) + 3f_{xyy}(x, y) + f_{yyy}(x, y)) \\ &\quad + \theta(h^4) \end{aligned}$$

When these are expanded using the coefficients A, B, C, D, E, F, G, we get the following terms:

$$\rightarrow (A + B + C + D + E + F + G) \cdot f(x, y)$$

$$\rightarrow (A + B - E - G)h \cdot f_x(x, y)$$

$$\rightarrow (A + C - F - G)h \cdot f_y(x, y)$$

$$\rightarrow (A + B + E + G)\frac{h^2}{2} \cdot f_{xx}(x, y)$$

$$\rightarrow (2A + 2G)\frac{h^2}{2} \cdot f_{xy}(x, y)$$

$$\rightarrow (A + C + F + G)\frac{h^2}{2} \cdot f_{yy}(x, y)$$

$$\rightarrow (A + B - E - G)\frac{h^3}{6} \cdot f_{xxx}(x, y)$$

$$\rightarrow (3A - 3G)\frac{h^3}{6} \cdot f_{xxy}(x, y)$$

$$\rightarrow (3A - 3G)\frac{h^3}{6} \cdot f_{xyy}(x, y)$$

$$\rightarrow (A + C - F - G)\frac{h^3}{6} \cdot f_{yyy}(x, y)$$

Now, we combine this with the fact that only a $f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$ term appears in the LHS.

We have now found 10 equations, but upon inspection, we find that 3 of these are repeated $\Rightarrow$ we have 7 equations and 7 unknowns

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & -1 & 0 & -1 \\ 1 & 0 & 1 & 0 & 0 & -1 & -1 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 3 & 0 & 0 & 0 & 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \\ G \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2h^2 \\ 0 \\ 0 \end{bmatrix}$$

Now, to solve this system:

$$\text{we see } 3A - 3G = 0 \Rightarrow A = G$$

$$\therefore 2A + 2G = 4A = \frac{2}{h^2} \Rightarrow A = G = \frac{1}{2h^2} \quad \checkmark$$

$$\text{Now, } A + C - F - G = 0 \Rightarrow C = F$$

$$\text{and } A + B - E - G = 0 \Rightarrow B = E$$

$$\therefore A + B + E + G = 2B + \frac{1}{h^2} = 0 \Rightarrow B = E = -\frac{1}{2h^2} \quad \checkmark$$

$$\therefore A + C + F + G = 2C + \frac{1}{h^2} = 0 \Rightarrow C = F = -\frac{1}{2h^2} \quad \checkmark$$

This leaves: $A + B + C + D + E + F + G = 0$

$$\therefore \frac{1}{2h^2} - \frac{1}{2h^2} - \frac{1}{2h^2} + D - \frac{1}{2h^2} - \frac{1}{2h^2} + \frac{1}{2h^2} = 0$$

$$\therefore D = \frac{1}{h^2} \quad \checkmark$$

$$\therefore \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \\ G \end{bmatrix} = \begin{bmatrix} 1/2h^2 \\ -1/2h^2 \\ -1/2h^2 \\ 1/h^2 \\ -1/2h^2 \\ -1/2h^2 \\ 1/2h^2 \end{bmatrix} = \frac{1}{2h^2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 2 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

$$\therefore \text{error term} = \mathcal{O}(|A|+|B|+|C|+|D|+|E|+|F|+|G|)h^4 \\ = \mathcal{O}(h^2)$$

Now, plugging these values in:

$$\frac{\partial^2 f}{\partial x \partial y} = Af(x+h, y+h) + Bf(x+h, y) + Cf(x, y+h) \\ + Df(x, y) + Ef(x-h, y) + Ff(x, y-h) \\ + Gf(x-h, y-h) + \mathcal{O}(h^2)$$

$$= \frac{1}{2h^2} f(x+h, y+h) - \frac{1}{2h^2} f(x+h, y) - \frac{1}{2h^2} f(x, y+h) \\ + \frac{2}{2h^2} f(x, y) - \frac{1}{2h^2} f(x-h, y) - \frac{1}{2h^2} f(x, y-h) \\ + \frac{1}{2h^2} f(x-h, y-h) + \mathcal{O}(h^2)$$

$$\therefore \frac{\partial^2 f}{\partial x \partial y} = \frac{1}{2h^2} \left(f(x+h, y+h) - f(x+h, y) - f(x, y+h) \right. \\ \left. + 2f(x, y) - f(x-h, y) - f(x, y-h) \right. \\ \left. + f(x-h, y-h) \right) + \mathcal{O}(h^2)$$



Q.E.D.

Problem #3

Given: $\int_{-1}^1 f(x) dx \approx c_1 f(x_1) + c_2 f(x_2) + c_3 f(x_3)$

$\therefore$ 6 unknowns: $c_1, c_2, c_3, x_1, x_2, x_3$

To find: use definition for exactness to come up with simultaneous equations.

6 unknowns $\Rightarrow$ 6 equations $\Rightarrow$ up to degree 5

- $f(x) = 1$
 $\hookrightarrow$ LHS = $\int_{-1}^1 dx = 2$

$\hookrightarrow$ RHS = $c_1 + c_2 + c_3$

Assuming exactness:

$$c_1 + c_2 + c_3 = 2 \quad \dots (i)$$

- $f(x) = x$
 $\hookrightarrow$ LHS = $\int_{-1}^1 x dx = 0$

$\hookrightarrow$ RHS = $c_1 x_1 + c_2 x_2 + c_3 x_3$

Assuming exactness:

$$c_1 x_1 + c_2 x_2 + c_3 x_3 = 0 \quad \dots (ii)$$

- $f(x) = x^2$
 $\hookrightarrow$ LHS = $\int_{-1}^1 x^2 dx = \frac{2}{3}$

$\hookrightarrow$ RHS = $c_1 x_1^2 + c_2 x_2^2 + c_3 x_3^2$

Assuming exactness:

$$c_1 x_1^2 + c_2 x_2^2 + c_3 x_3^2 = \frac{2}{3} \quad \dots (iii)$$

- $f(x) = x^3$
 $\hookrightarrow \text{LHS} = \int_{-1}^1 x^3 dx = 0$

$$\hookrightarrow \text{RHS} = c_1 x_1^3 + c_2 x_2^3 + c_3 x_3^3$$

Assuming exactness:

$$c_1 x_1^3 + c_2 x_2^3 + c_3 x_3^3 = 0 \quad \dots \quad (\text{iv})$$

- $f(x) = x^4$
 $\hookrightarrow \text{LHS} = \int_{-1}^1 x^4 dx = \frac{2}{5}$

$$\hookrightarrow \text{RHS} = c_1 x_1^4 + c_2 x_2^4 + c_3 x_3^4$$

Assuming exactness:

$$c_1 x_1^4 + c_2 x_2^4 + c_3 x_3^4 = \frac{2}{5} \quad \dots \quad (\text{v})$$

- $f(x) = x^5$
 $\hookrightarrow \text{LHS} = \int_{-1}^1 x^5 dx = 0$

$$\hookrightarrow \text{RHS} = c_1 x_1^5 + c_2 x_2^5 + c_3 x_3^5$$

Assuming exactness:

$$c_1 x_1^5 + c_2 x_2^5 + c_3 x_3^5 = 0 \quad \dots \quad (\text{vi})$$

Thus, we have the 6 equations:

$$\left\{ \begin{array}{l} c_1 + c_2 + c_3 = 2 \\ c_1 x_1 + c_2 x_2 + c_3 x_3 = 0 \\ c_1 x_1^2 + c_2 x_2^2 + c_3 x_3^2 = 2/3 \\ c_1 x_1^3 + c_2 x_2^3 + c_3 x_3^3 = 0 \\ c_1 x_1^4 + c_2 x_2^4 + c_3 x_3^4 = 2/5 \\ c_1 x_1^5 + c_2 x_2^5 + c_3 x_3^5 = 0 \end{array} \right.$$

When this is solved using a symbolic calculator, we get the following solutions:

- $c_1 = 0, c_2 = 1, c_3 = 1, x_1 = -\frac{\sqrt[4]{13/5}}{\sqrt{3}}, x_2 = -\frac{1}{\sqrt{3}}, x_3 = \frac{1}{\sqrt{3}}$
- $c_1 = 0, c_2 = 1, c_3 = 1, x_1 = -\frac{\sqrt[4]{13/5}}{\sqrt{3}}, x_2 = \frac{1}{\sqrt{3}}, x_3 = -\frac{1}{\sqrt{3}}$
- $c_1 = 0, c_2 = 1, c_3 = 1, x_1 = -\frac{i\sqrt[4]{13/5}}{\sqrt{3}}, x_2 = -\frac{1}{\sqrt{3}}, x_3 = \frac{1}{\sqrt{3}}$
- $c_1 = 0, c_2 = 1, c_3 = 1, x_1 = -\frac{i\sqrt[4]{13/5}}{\sqrt{3}}, x_2 = \frac{1}{\sqrt{3}}, x_3 = -\frac{1}{\sqrt{3}}$
- $c_1 = 0, c_2 = 1, c_3 = 1, x_1 = \frac{i\sqrt[4]{13/5}}{\sqrt{3}}, x_2 = -\frac{1}{\sqrt{3}}, x_3 = \frac{1}{\sqrt{3}}$

Now we can notice that:

- c_1 corresponds to $f(x_1)$, but $c_1 = 0$ in all the solutions, so the quadrature rule only depends on $c_2 f(x_2)$ and $c_3 f(x_3)$
- $c_2 = 1$ and $c_3 = 1$ always. In all the solutions $x_3 = -x_2$, so by the symmetry of the coefficients c , it does not matter which of the possible values is assigned to x_2 and x_3 .

$$\therefore \text{ we get: } \int_{-1}^1 f(x) dx \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

Due to its derivation, this quadrature is exact up to degree - 5.

Now, we can test for higher degrees:

$$\bullet f(x) = x^6 \quad \int_{-1}^1 x^6 dx = \frac{2}{7}$$

$$\hookrightarrow \text{RHS} = \left(-\frac{1}{\sqrt{3}}\right)^6 + \left(\frac{1}{\sqrt{3}}\right)^6 = \frac{1}{2^7} + \frac{1}{2^7} = \frac{2}{2^7} \neq \frac{2}{7}$$

$\therefore$ not exact for degree 6.

∴ we obtain the quadrature:

$$\int_{-1}^1 f(x) dx \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

which is exact up to degree 5.

Problem #4:

$$a) \quad A = \begin{bmatrix} 10 & -1 & 6 \\ -1 & 8 & 9 \\ 6 & 9 & 1 \end{bmatrix}, \quad \vec{v}^{(0)} = \begin{bmatrix} 0.5 \\ 0.6 \\ 0.6 \end{bmatrix}$$

The power iteration algorithm is:

$$\begin{cases} \text{for } k=1, 2, \dots, n: \\ \quad w = A \vec{v}^{(k-1)} \\ \quad \vec{v}^{(k)} = \frac{w}{\|w\|_2} \end{cases}$$

Using MATLAB to perform these iterations gives:

$$\rightarrow \vec{v}^{(2)} = \begin{bmatrix} 0.511937 \\ 0.620256 \\ 0.594309 \end{bmatrix} = \begin{bmatrix} 0.5119 \\ 0.6203 \\ 0.5943 \end{bmatrix} \quad (= \vec{v}_1)$$

To find $\lambda^{(2)}$, we have the formula for Rayleigh's Quotient:

$$r(\vec{x}) = \frac{\vec{x}^T A \vec{x}}{\|\vec{x}\|_2}$$

$\lambda^{(2)} = r(\vec{v}^{(2)})$. Using MATLAB, this gives:

$$\begin{aligned} \lambda^{(2)} &= 15.702884 \\ \therefore \lambda_1 &= 15.7029 \end{aligned}$$

$$b) \text{ We have: } \vec{v}^{(2)} = \begin{bmatrix} 0.5119 \\ 0.6203 \\ 0.5943 \end{bmatrix}, \quad \lambda^{(2)} = 15.7029$$

$\therefore$ the relative error, using MATLAB, is:

$$\begin{aligned} \frac{\|A \vec{v}^{(2)} - \lambda^{(2)} \vec{v}^{(2)}\|_2}{\|\lambda^{(2)} \vec{v}^{(2)}\|_2} &= 0.006753993 \\ &= 0.0068 \end{aligned}$$

c) To deflate the matrix: $A' = A - \lambda_1 \vec{p}_1 \vec{p}_1^T$

We have $\lambda_1 = \lambda^{(1)}$ and $\vec{p}_1 = \vec{v}^{(1)}$. Using MATLAB, we get:

$$A' = \begin{bmatrix} 5.8846 & -5.9862 & 1.2224 \\ -5.9862 & 1.9588 & 3.2116 \\ 1.2224 & 3.2116 & -4.5463 \end{bmatrix}$$

$$d) A' = \begin{bmatrix} 5.8846 & -5.9862 & 1.2224 \\ -5.9862 & 1.9588 & 3.2116 \\ 1.2224 & 3.2116 & -4.5463 \end{bmatrix}, \vec{v}^{(1)} = \begin{bmatrix} 0.5 \\ 0.6 \\ 0.6 \end{bmatrix}$$

I put this system into MATLAB and used the same iteration procedure as described above in Part (a).

After 100 iterations (to ensure convergence, and for redundancy), we get:

$$\vec{v}^{(100)} = \begin{bmatrix} 0.7982 \\ -0.5990 \\ -0.0639 \end{bmatrix}$$

Now, $\lambda^{(100)} = \mu(\vec{v}^{(100)})$:

$$\begin{aligned} \lambda^{(100)} &= 10.278788517 \\ &= 10.2788 \end{aligned}$$

$$\Rightarrow \lambda_2 = 10.2788 \longleftrightarrow \vec{v}_2 = \begin{bmatrix} 0.7982 \\ -0.5990 \\ -0.0639 \end{bmatrix}$$

e) λ and $\vec{v}$ obtained from (d) are an eigenvalue-eigenvector pair:

$$A\vec{v}_2 \approx \lambda_2 \vec{v}_2$$